
Problem 8.17. Let A be the language of properly nested parentheses. For example, (()) and (()(()))() are in A, but)(is not. Show that A is in L.

Proof. To show that $A \in \mathrm{L}$, we give a log space algorithm M that decides A.

$M =$ "On input $\langle w \rangle$, where w is a string:

1. Let c be a binary integer initialized with 0.
2. Repeat for each symbol $b = w_1, w_2, \ldots, w_n$.
3. If b is (, then increment c by 1.
4. If b is) and $c > 0$, then increment c by 1.
5. If b is) and $c = 0$, then *reject*.
6. If $c = 0$, then *accept*. Otherwise, *reject*."

$\square$